

# How Accurate Is the Voice Emotion Model?

AI Actor Coach (ActorPro AI) · Final Year Project · October 2026

## 1. What the model does

When a student says a line, the app sends the recording to its voice service. The voice emotion model listens to **how** the line was said (tone, energy, pitch), not the words, and gives a percentage for each of seven emotions: anger, disgust, fear, joy, neutral, sadness and surprise. The model is **emotion2vec+ large** (about 300 million parameters). Before it listens, the app cuts the recording down to just the speech, and the percentages are adjusted so they match how sure the model really is.

## 2. Datasets we tested on

We ran the model on two labelled datasets of emotional speech. Every clip went through exactly the same steps as a recording in the app. Neither dataset's people were used to build the app.

### CREMA-D Acted speech · clear, deliberate emotions

|                |                                                                                                                                   |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Type of voices | 91 actors (48 male, 43 female, aged 20 to 74, of several ethnic backgrounds) reading short sentences with an emotion.             |
| Conditions     | Recorded in a studio: clean sound, no background noise.                                                                           |
| What is said   | 12 everyday sentences such as "Don't forget a jacket" and "It's eleven o'clock". The words carry no emotion, only the voice does. |
| Emotions       | Anger, disgust, fear, joy, neutral, sadness (no surprise).                                                                        |
| Full dataset   | 7,442 recordings.                                                                                                                 |
| What we used   | <b>600 clips</b> : 100 for each emotion, picked at random.                                                                        |
| Why it matters | Closest to a student deliberately acting an emotion.                                                                              |

### MELD TV dialogue · natural, everyday emotion

|                |                                                                                                                  |
|----------------|------------------------------------------------------------------------------------------------------------------|
| Type of voices | The cast of the TV show <i>Friends</i> in real scenes: natural conversation, interruptions and overlapping talk. |
| Conditions     | TV audio with background music, laughter and noise.                                                              |
| What is said   | Real dialogue lines, often very short ("Oh my God!", "What?").                                                   |
| Emotions       | All seven: anger, disgust, fear, joy, neutral, sadness, surprise.                                                |
| Full dataset   | About 13,000 lines; the official test part has 2,610.                                                            |
| What we used   | <b>618 clips</b> from the test part: up to 100 per emotion (only 68 disgust and 50 fear exist).                  |
| Why it matters | Real, subtle emotion in noisy audio: a much harder test, closer to a casual home recording.                      |

## 3. Results

| Dataset            | Clips tested | Overall accuracy | Balanced accuracy (95% range) | Random guessing |
|--------------------|--------------|------------------|-------------------------------|-----------------|
| CREMA-D (acted)    | 595          | <b>84.9%</b>     | <b>84.8%</b> (82%–88%)        | 17%             |
| MELD (TV dialogue) | 576          | <b>34.4%</b>     | <b>31.0%</b> (28%–34%)        | 14%             |

*Overall accuracy* = share of clips where the model's top emotion was the right one. *Balanced accuracy* counts every emotion equally, so the common ones cannot hide the rare ones. Clips with too little speech were skipped, as the app does (47 in total). Average time: 0.18 s per clip on a laptop CPU.

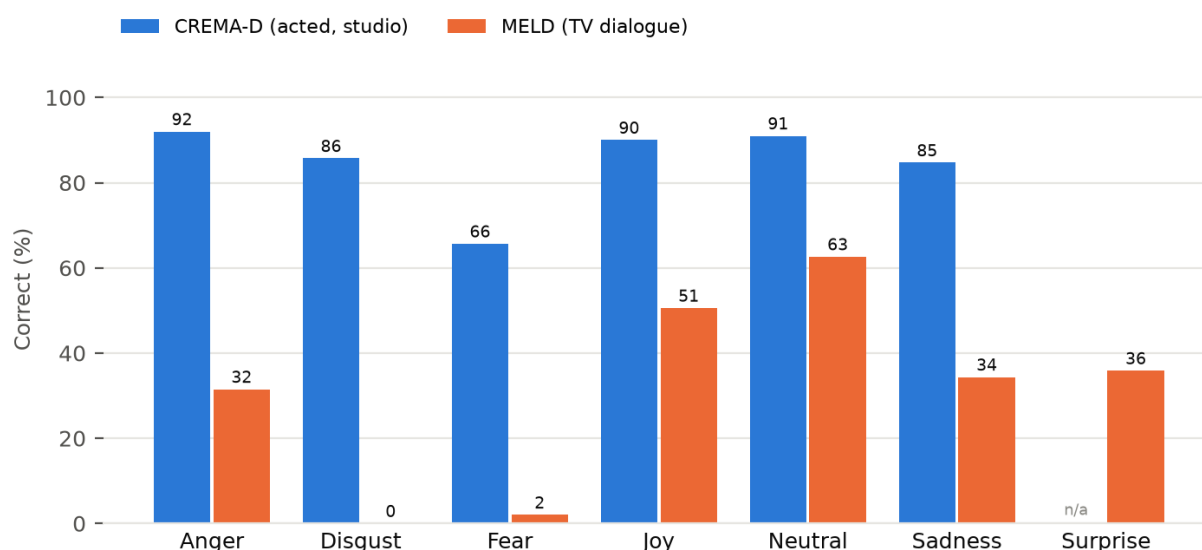


Figure 1. How often each emotion was recognised correctly. CREMA-D has no surprise clips.

## 4. Which emotions get mixed up

Each row is the real emotion; each column is what the model heard. Blue = correct, orange = a common mistake (15% or more).

### CREMA-D (acted)

| Actual \ Heard as | Anger | Disgust | Fear | Joy | Neutral | Sadness |
|-------------------|-------|---------|------|-----|---------|---------|
| Anger             | 92%   | 6%      | 1%   | 1%  | —       | —       |
| Disgust           | 1%    | 86%     | 3%   | 1%  | —       | 9%      |
| Fear              | 2%    | 3%      | 66%  | 3%  | 3%      | 23%     |
| Joy               | 2%    | 4%      | 1%   | 90% | 3%      | —       |
| Neutral           | —     | —       | 1%   | 3%  | 91%     | 5%      |
| Sadness           | 1%    | 4%      | 6%   | 1%  | 3%      | 85%     |

### MELD (TV dialogue)

| Actual \ Heard as | Anger | Disgust | Fear | Joy | Neutral | Sadness | Surprise |
|-------------------|-------|---------|------|-----|---------|---------|----------|
| Anger             | 32%   | —       | 1%   | 14% | 33%     | 8%      | 13%      |
| Disgust           | 21%   | —       | 2%   | 26% | 32%     | 9%      | 11%      |
| Fear              | 10%   | —       | 2%   | 26% | 28%     | 16%     | 18%      |
| Joy               | 3%    | —       | 1%   | 51% | 30%     | 6%      | 9%       |
| Neutral           | 2%    | —       | —    | 24% | 63%     | 4%      | 7%       |
| Sadness           | 4%    | —       | —    | 18% | 32%     | 34%     | 11%      |
| Surprise          | 8%    | 1%      | —    | 26% | 21%     | 9%      | 36%      |

- **Fear** was heard as **sadness** 23% of the time (CREMA-D).
- **Anger** was heard as **neutral** 33% of the time (MELD).
- **Sadness** was heard as **neutral** 32% of the time (MELD).
- **Disgust** was heard as **neutral** 32% of the time (MELD).

- **Joy** was heard as **neutral** 30% of the time (MELD).

## 5. What this means

- **On clearly acted emotion the model is strong:** 85% of CREMA-D clips were recognised correctly, against 17% for random guessing. Its best emotion there was anger (92%), its weakest fear (66%).
- **On natural TV dialogue it is much weaker:** 31% balanced accuracy on MELD (chance 14%). MELD is famously hard: short lines, background laughter and subtle emotion. Its best emotion there was neutral (63%), its weakest disgust (0%).
- **It rarely mistakes acted emotion for neutral:** only 2% of emotional CREMA-D clips were called neutral (29% on MELD), so clear acting is usually rewarded.
- **For students:** the clearer and more committed the delivery, the more reliable the result. Subtle or very short lines are the hardest to judge.

## 6. Honest limits

- The model's creators trained it partly on a large collection of public emotion datasets, which may include CREMA-D and MELD's training part. The CREMA-D result may therefore be somewhat optimistic; MELD's official test part is the fairer number.
- Both datasets are in English with mostly North American accents. Accuracy for other accents and laptop microphones has not been measured yet; the app can save a student's own takes (on their own computer) to check this.
- These are samples (600 and 618 clips), so the numbers have a margin of error; the 95% ranges are shown in Section 3.

**Credits:** emotion2vec+ (Ma et al., ACL 2024; FunASR model licence). CREMA-D (Cao et al., 2014; Open Database License). MELD (Poria et al., ACL 2019; GPL-3.0).